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TDD-Mockito

Exercise 1:Implementing the Singleton Pattern

Code:

```
import org.junit.jupiter.api.Test;

import org.mockito.Mockito;

import static org.mockito.Mockito.*;

import static org.junit.jupiter.api.Assertions.*;

public class MockingStubbingTest {

    @Test

    public void testExternalApi() {

        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        when(mockApi.getData()).thenReturn("Mock Data")

        MyService service = new MyService(mockApi);

        assertEquals("Mock Data", result);

    }

    @Test

    public void testStubWithQuery() {

        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        when(mockApi.fetchData("test")).thenReturn("Stubbed Result");
```

```
MyService service = new MyService(mockApi);

String result = service.fetchData("test");

assertEquals("Stubbed Result", result);

}
```

@Test

```
public void testStubMultipleCalls() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    // Stub to return different values on consecutive calls
    when(mockApi.getData())

        .thenReturn("First Call")

        .thenReturn("Second Call");

    MyService service = new MyService(mockApi);

    assertEquals("First Call", service.fetchData());

    assertEquals("Second Call", service.fetchData());

}
```

@Test

```
public void testStubWithException() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    when(mockApi.getData()).thenThrow(new RuntimeException("API
Error"));

    MyService service = new MyService(mockApi);

    assertThrows(RuntimeException.class, () -> {

        service.fetchData();

    });

}
```

Exercise 2: Implementing the Factory Method Pattern

code :

```
import org.junit.jupiter.api.Test;

import org.mockito.Mockito;

import static org.mockito.Mockito.*;

import static org.junit.jupiter.api.Assertions.*;

public class VerifyInteractionTest {

    @Test

    public void testVerifyInteraction() {

        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        MyService service = new MyService(mockApi);

        service.fetchData();

        verify(mockApi).getData();

    }

    @Test

    public void testVerifyWithArguments() {

        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        MyService service = new MyService(mockApi);

        service.fetchData("testQuery");

        verify(mockApi).fetchData("testQuery");

    }

    @Test
```

```
public void testVerifyCallCount() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    MyService service = new MyService(mockApi);

    service.fetchData();

    service.fetchData();

    service.fetchData();

    verify(mockApi, times(3)).getData();

}

@Test

public void testVerifyNoInteractions() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    MyService service = new MyService(mockApi);

    verifyNoInteractions(mockApi);

}

@Test

public void testVerifyNeverCalled() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    MyService service = new MyService(mockApi);

    service.fetchData();

    verify(mockApi, never()).processData(anyString());

}

}
```

Setting Up JUnit

Code:

```
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class JUnitSetup {

    @Test

    public void testAddition() {

        int result = 2 + 3;

        assertEquals(5, result);

    }

    @Test

    public void testStringConcatenation() {

        String greeting = "Hello" + " " + "World";

        assertEquals("Hello World", greeting);

    }

    @Test

    public void testArrayLength() {

        int[] numbers = {1, 2, 3, 4, 5};

        assertEquals(5, numbers.length);

    }

}
```

OUTPUT:

```
java -Dnet.bytebuddy.experimental=true -jar  
lib/junit-platform-console-standalone-1.10.2.jar --class-path ".:lib/*" --select-class  
VerifyInteractionTest
```

♥ Thanks for using JUnit! Support its development at <https://junit.org/sponsoring>

```
|  
├─ JUnit Jupiter ✓  
│   └─ VerifyInteractionTest ✓  
│       ├── testVerifyInteraction() ✗ org/mockito/Mockito  
│       ├── testVerifyCallCount() ✗ org/mockito/Mockito  
│       ├── testVerifyWithArguments() ✗ org/mockito/Mockito  
│       ├── testVerifyNeverCalled() ✗ org/mockito/Mockito  
│       └─ testVerifyNoInteractions() ✗ org/mockito/Mockito  
├─ JUnit Vintage ✓  
└─ JUnit Platform Suite ✓
```

Failures (5):

```
JUnit Jupiter:VerifyInteractionTest:testVerifyInteraction()  
  MethodSource [className = 'VerifyInteractionTest', methodName =  
'testVerifyInteraction', methodParameterTypes = '']  
=> java.lang.NoClassDefFoundError: org/mockito/Mockito  
  VerifyInteractionTest.testVerifyInteraction(VerifyInteractionTest.java:14)  
  java.base/java.lang.reflect.Method.invoke(Method.java:580)  
  java.base/java.util.ArrayList.forEach(ArrayList.java:1597)  
  java.base/java.util.ArrayList.forEach(ArrayList.java:1597)  
Caused by: java.lang.ClassNotFoundException: org.mockito.Mockito  
  java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:445)  
  java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:595)  
  java.base/java.net.FactoryURLClassLoader.loadClass(URLClassLoader.java:870)  
  java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:528)  
  [...]  
JUnit Jupiter:VerifyInteractionTest:testVerifyCallCount()  
  MethodSource [className = 'VerifyInteractionTest', methodName =  
'testVerifyCallCount', methodParameterTypes = '']  
=> java.lang.NoClassDefFoundError: org/mockito/Mockito  
  VerifyInteractionTest.testVerifyCallCount(VerifyInteractionTest.java:44)  
  java.base/java.lang.reflect.Method.invoke(Method.java:580)  
  java.base/java.util.ArrayList.forEach(ArrayList.java:1597)  
  java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
```

```
Caused by: java.lang.ClassNotFoundException: org.mockito.Mockito
[...]
JUnit Jupiter:VerifyInteractionTest:testVerifyWithArguments()
MethodSource [className = 'VerifyInteractionTest', methodName =
'testVerifyWithArguments', methodParameterTypes = '']
=> java.lang.NoClassDefFoundError: org/mockito/Mockito
VerifyInteractionTest.testVerifyWithArguments(VerifyInteractionTest.java:29)
java.base/java.lang.reflect.Method.invoke(Method.java:580)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
Caused by: java.lang.ClassNotFoundException: org.mockito.Mockito
[...]
JUnit Jupiter:VerifyInteractionTest:testVerifyNeverCalled()
MethodSource [className = 'VerifyInteractionTest', methodName =
'testVerifyNeverCalled', methodParameterTypes = '']
=> java.lang.NoClassDefFoundError: org/mockito/Mockito
VerifyInteractionTest.testVerifyNeverCalled(VerifyInteractionTest.java:73)
java.base/java.lang.reflect.Method.invoke(Method.java:580)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
Caused by: java.lang.ClassNotFoundException: org.mockito.Mockito
[...]
JUnit Jupiter:VerifyInteractionTest:testVerifyNoInteractions()
MethodSource [className = 'VerifyInteractionTest', methodName =
'testVerifyNoInteractions', methodParameterTypes = '']
=> java.lang.NoClassDefFoundError: org/mockito/Mockito
VerifyInteractionTest.testVerifyNoInteractions(VerifyInteractionTest.java:61)
java.base/java.lang.reflect.Method.invoke(Method.java:580)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
java.base/java.util.ArrayList.forEach(ArrayList.java:1597)
Caused by: java.lang.ClassNotFoundException: org.mockito.Mockito
[...]
```

Test run finished after 69 ms

```
[ 4 containers found ]
[ 0 containers skipped ]
[ 4 containers started ]
[ 0 containers aborted ]
[ 4 containers successful ]
[ 0 containers failed ]
[ 5 tests found ]
```

```
[ 0 tests skipped ]
[ 5 tests started  ]
[ 0 tests aborted  ]
[ 0 tests successful ]
[ 5 tests failed   ]
```

WARNING: Delegated to the 'execute' command.

This behaviour has been deprecated and will be removed in a future release.

Please use the 'execute' command directly.